

CHAPTER 3 Epidemiologic Measurements Used to Describe Disease Occurrence

Specific Mortality Rates

A **specific mortality rate** refers to a mortality rate in a particular subgroup of the population defined in terms of race, age, or sex. A specific mortality rate also may refer to the entire population but for some particular cause of death. The three examples of specific mortality rates discussed in this chapter are cause-specific mortality rates, age-specific mortality rates, and sex-specific mortality rates. You will learn how they can be applied in various situations.

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The *cause-specific mortality rate* is a measure that refers to mortality from a specific cause divided by the population size at the midpoint of a time period. The formula for the cause-specific mortality rate is shown in the text box. An important consideration when calculating a cause-specific mortality rate is that anyone included in the population in the denominator must be able to die from that specific cause. For example, a testicular cancer specific mortality rate would only include males in the denominator.

$$= \frac{\text{Number of deaths from a specific cause}}{\text{Population size at midpoint of time period}} \times multiplier$$

- Time period needs to be specified.
- Multiplier is usually either 1,000 or 100,000.
- *Calculation example:* Refer to Table 3.4 for the data used in this calculation.
 - 200,955 deaths from accidents (unintentional injuries) in 2020
 - Population as of July 1, 2020 = 331,449,520

$$= \frac{200,955}{331,449.520} \times 100,000$$

- $= 60.6 \text{ per } 100,000$

Age-Specific Mortality Rates

An *age-specific mortality rate* refers to the number of deaths in an age group of the population during a specified time period. Age-specific mortality rates help in making comparisons regarding a cause of mortality across age groups. The formula for an age-specific mortality rate is shown in the text box.

Age-Specific Mortality Rate

$$= \frac{\text{Number of deaths among those in an age group}}{\text{Population size in that age group}} \times \text{multiplier}$$

- Time period needs to be specified.
 - Multiplier is usually either 1,000 or 100,000.
 - *Calculation example:*
In the United States in 2020 for the age group 15 to 24 years,
 - 35,470 deaths from all causes
 - Estimated population in that age group = 44,000,000
- Age-specific mortality rate**
- $$= \frac{35,470}{44,000,000} \times 100,000 = 80.6 \text{ per } 100,000$$

Figure 3.11 illustrates age-specific firearm-related mortality rates in the United States in 2020. Mortality rates are highest for the 20 to 24 year age group and decrease with increasing age.

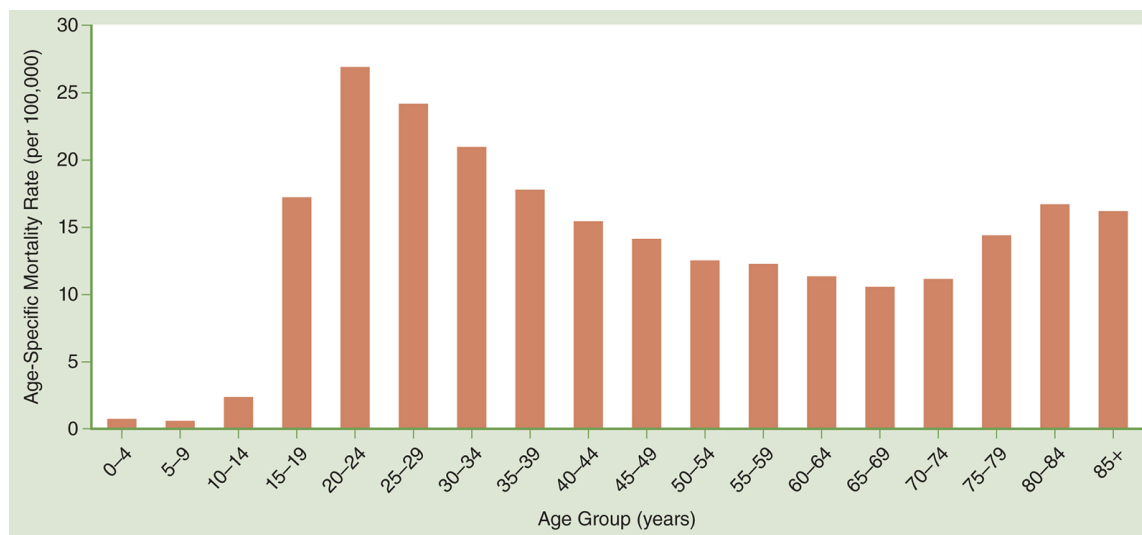


Figure 3.11 Age-specific firearm mortality rates per 100,000, United States, 2020

Data from Centers for Disease Control and Prevention National Centers Injury Prevention and Control. Web-based Injury Statistics Query and Reporting System (WISQARS). Available at:
www.cdc.gov/injury/wisqars.
 Description

Sex-Specific Mortality Rates

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A sex-specific mortality rate refers to the number of deaths in a sex group divided by the total number of persons within that sex group during a time period. The formula for a sex-specific mortality rate is shown in the text box.

Sex-Specific Mortality Rate

$$= \frac{\text{Number of deaths in a sex group}}{\text{Population size in that sex group}} \times \text{multiplier}$$

- Time period needs to be specified.
- Multiplier is usually either 1,000 or 100,000.
- *Calculation example:*
 - In the United States in 2020,
 - 1,691,900 deaths among males
 - 1,547,093 deaths among females
 - Estimated number of males in the population as of July 1, 2020 = 164,067,512
 - Estimated number of females in the population as of July 1, 2020 = 167,382,008
 - Male-specific mortality rate
 - $= \frac{1,691,900}{164,067,512} \times 100,000 = 1,031.2 \text{ per } 100,000$
 - Female-specific mortality rate
 - $= \frac{1,547,093}{167,382,008} \times 100,000 = 924.3 \text{ per } 100,000$